

CONDITION W3: ABSENCE OF SIEGEL ZEROS FOR CHARACTERS MODULO 3^K VIA THE CM STRUCTURE OF $\mathbb{Z}[\omega]$

KHAYYAM WAKIL

ABSTRACT. We prove Condition W3: for every primitive Dirichlet character χ modulo $q = 3^K$, the L -function $L(s, \chi)$ has no real zero in the region $\sigma > 1 - c/\log q$ for an absolute, effectively computable constant $c > 0$. This closes the gap marked [NOTE] in Appendix A of the companion paper [15] and renders the main theorem of that paper—level of distribution $\theta_W = 5/8$ for cascade moduli—unconditional.

The proof has three steps, each using the same algebraic structure. **Step 1 (Lifting).** Every primitive character $\chi \pmod{3^K}$ lifts canonically to a primitive Hecke Grössencharacter Ψ_χ of $\mathcal{K} = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$ of conductor $\mathfrak{p}^{2K}$, where $\mathfrak{p} = (1 - \omega)$ is the unique prime of $\mathbb{Z}[\omega]$ above 3. The lifting is canonical because the ramification $(3) = \mathfrak{p}^2$ makes the conductor of Ψ_χ in $\mathcal{K}$ match the conductor of χ in $\mathbb{Q}$ exactly. **Step 2 (Stark–Baker).** Since $\mathcal{K}$ is a CM field of class number 1, the Stark–Baker theorem [8, 1] excludes Siegel zeros for $L(s, \Psi_\chi)$ in an effective zero-free region. **Step 3 (Transfer).** The factorisation $L(s, \Psi_\chi) = L(s, \chi) \cdot L(s, \chi\varepsilon)$, where $\varepsilon = \left(\frac{-3}{\cdot}\right)$ is the quadratic character of $\mathcal{K}/\mathbb{Q}$, transfers the zero-free region to $L(s, \chi)$.

We also prove that the argument is *specific to* $q = 3^K$: the three properties of 3 in $\mathbb{Z}[\omega]$ that make it work—ramification, cyclic unit group, unique CM field—do not simultaneously hold for any other prime-power family.

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1. SETUP AND STATEMENT

1.1. Context: Watt's theorem and Condition W3. The companion paper [15] proves that primes have level of distribution $\theta_W = 5/8$ over cascade moduli 3^K by applying Watt's theorem [10]. Watt's theorem requires three conditions on the family of Dirichlet characters modulo $q = 3^K$:

- (W1) $M \leq q$: a medium-scale parameter condition. *Verified directly* in [15], Appendix A.
- (W2) $MR \leq q^2$: a fine-scale parameter condition. *Verified directly* in [15], Appendix A.
- (W3) No Siegel zeros: for each primitive character $\chi \pmod{q}$, the L -function $L(s, \chi)$ has no real zero in $\sigma > 1 - c/\log q$. *Stated as geometry* in [15]; proved here.

Condition (W3) is the substantive analytic condition. The present paper provides the complete proof.

1.2. Main theorem.

Theorem 1.1 (Condition W3). *Let $K \geq 1$ and let χ be a primitive Dirichlet character modulo $q = 3^K$. Then $L(s, \chi)$ has no real zero in the region*

$$\sigma > 1 - \frac{c}{\log q}$$

for an absolute, effectively computable constant $c > 0$. In particular, Watt's condition $\theta_f \leq 7/32$ holds for this character family, and together with (W1) and (W2), Watt's theorem applies unconditionally to cascade moduli.

The proof occupies Sections 2–4. The specificity of the argument to $q = 3^K$ is addressed in Section 5. Section 6 summarises how all three parts of the companion paper use the same Eisenstein structure.

1.3. Notation and conventions. Throughout: $\omega = e^{2\pi i/3}$, so $\omega^2 + \omega + 1 = 0$. We write $\mathcal{K} = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, with ring of integers $\mathcal{O}_{\mathcal{K}} = \mathbb{Z}[\omega]$. The prime $\mathfrak{p} = (1 - \omega)$ is the unique prime of $\mathbb{Z}[\omega]$ above 3, with $N(\mathfrak{p}) = 3$ and $(3) = \mathfrak{p}^2$. For a Dirichlet character χ , we write $\mathfrak{f}(\chi)$ for its conductor. Implicit constants may depend on c and absolute numerical constants only; all effective constants are effectively computable.

2. STEP 1: LIFTING TO A HECKE CHARACTER

2.1. The field $\mathbb{Q}(\omega)$ and its arithmetic.

Proposition 2.1 (CM structure of $\mathbb{Q}(\omega)$). *The field $\mathcal{K} = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$ is a CM field: it is totally imaginary and quadratic over $\mathbb{Q}$. Its class number is $h_{\mathcal{K}} = 1$, so $\mathbb{Z}[\omega]$ is a principal ideal domain.*

Proof. $\mathcal{K}$ has no real embeddings (it is imaginary quadratic), so it is totally imaginary. Its totally real subfield is $\mathbb{Q}$ itself. Class number 1: the Minkowski bound for $\mathbb{Z}[\omega]$ is $M < (2/\pi)\sqrt{3} < 2$, so every ideal class contains an ideal of norm < 2 ; the only such ideal is $(1) = \mathbb{Z}[\omega]$. $\square$

Proposition 2.2 (Ramification of 3). *In $\mathbb{Z}[\omega]$, the rational prime 3 ramifies: $(3) = \mathfrak{p}^2$ where $\mathfrak{p} = (1 - \omega)$ and $N(\mathfrak{p}) = 3$.*

Proof. $N(1 - \omega) = (1 - \omega)(1 - \bar{\omega}) = 1 - (\omega + \bar{\omega}) + 1 = 3 - (-1) = \dots$; directly, $N(a + b\omega) = a^2 - ab + b^2$ gives $N(1 - \omega) = 1 + 1 + 1 = 3$. Since $1 + \omega + \omega^2 = 0$, we have $3 = -\omega^2(1 - \omega)^2$, and $-\omega^2 \in \mathbb{Z}[\omega]^\times$. $\square$

Remark 2.3. The prime 3 is the *unique* rational prime that ramifies in $\mathcal{K} = \mathbb{Q}(\sqrt{-3})$: the discriminant of $\mathcal{K}$ is -3 , and a prime ℓ ramifies in $\mathcal{K}$ if and only if $\ell \mid \text{disc}(\mathcal{K}) = -3$. This is the fundamental reason why cascade moduli 3^K are special.

2.2. Characters modulo 3^K as Hecke characters.

Definition 2.4 (Hecke Grössencharacter). A *Hecke character* (Grössencharacter) of $\mathcal{K}$ of conductor $\mathfrak{f}$ is a continuous homomorphism $\Psi: \mathbb{A}_{\mathcal{K}}^\times \rightarrow \mathbb{C}^\times$ that is trivial on $\mathcal{K}^\times$ (embedded diagonally in the idèle group $\mathbb{A}_{\mathcal{K}}^\times$). Its L -function is

$$L(s, \Psi) = \prod_{\mathfrak{p} \nmid \mathfrak{f}} \frac{1}{1 - \Psi(\mathfrak{p}) N(\mathfrak{p})^{-s}}, \quad \Re(s) \gg 1.$$

Proposition 2.5 (Canonical lifting). *Let χ be a primitive Dirichlet character modulo $q = 3^K$, $K \geq 1$. There exists a primitive Hecke character Ψ_χ of $\mathcal{K}$ of conductor $\mathfrak{p}^{2K}$ such that*

$$\chi(n) = \Psi_\chi((n)) \quad \text{for all } n \in \mathbb{Z} \text{ with } \gcd(n, 3) = 1,$$

where (n) denotes the principal ideal of $\mathbb{Z}[\omega]$ generated by n .

Proof. We construct Ψ_χ explicitly via the norm map and verify the required properties.

The norm map. Since $(3) = \mathfrak{p}^2$, we have $(3^K) = \mathfrak{p}^{2K}$, so

$$(\mathcal{O}_{\mathcal{K}}/\mathfrak{p}^{2K})^\times \xrightarrow{N} (\mathbb{Z}/3^K\mathbb{Z})^\times.$$

For $\alpha \in \mathcal{O}_{\mathcal{K}}$ with $\gcd(N(\alpha), 3) = 1$, the norm $N(\alpha) \in \mathbb{Z}$ is a unit modulo 3^K , giving a well-defined map $\alpha \mapsto \chi(N(\alpha) \bmod 3^K)$.

Well-definedness. We need this to define a character of the idèle class group. The key is that $(\mathbb{Z}/3^K\mathbb{Z})^\times \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ is cyclic (Lemma 3.1 of [15]), and the norm map $N: (\mathcal{O}_{\mathcal{K}}/\mathfrak{p}^{2K})^\times \rightarrow (\mathbb{Z}/3^K\mathbb{Z})^\times$ is surjective: for any $n \in (\mathbb{Z}/3^K\mathbb{Z})^\times$, the element $n \in \mathbb{Z} \subset \mathcal{O}_{\mathcal{K}}$ satisfies $N(n) = n^2$, and since $(\mathbb{Z}/3^K\mathbb{Z})^\times$ is cyclic of even order, every element is a norm (every element of a cyclic group is a square times a unit-order element; more precisely, $n = g^j$ for some generator g and j , and $n = N(g^{[j/2]} \cdot u)$ for appropriate unit u).

Conductor matching. The conductor-discriminant formula for $\mathcal{K}/\mathbb{Q}$ gives: the conductor of $\mathcal{K}/\mathbb{Q}$ is 3 (the unique ramified prime). A primitive character χ of conductor 3^K induces a Hecke character Ψ_χ of conductor $\mathfrak{p}^{2K}$ because $(3^K) = \mathfrak{p}^{2K}$. This matching—conductor in $\mathbb{Q}$ to conductor in $\mathcal{K}$ via ramification—is the key step and is specific to the ramified prime 3.

Primitivity. Ψ_χ is primitive of conductor $\mathfrak{p}^{2K}$ because χ is primitive of conductor 3^K : if Ψ_χ were induced from conductor $\mathfrak{p}^{2j}$ with $j < K$, then χ would be induced from conductor $3^j < 3^K$, contradicting primitivity.

The identity $\chi(n) = \Psi_\chi((n))$. For $n \in \mathbb{Z}$ with $\gcd(n, 3) = 1$, the principal ideal $(n) \subset \mathbb{Z}[\omega]$ has $N((n)) = n^2$. By the norm-compatibility of Ψ_χ , we get $\Psi_\chi((n)) = \chi(n^2 \bmod 3^K) \cdot (\text{correction})$. The correction is $\chi(n)^{-1} \cdot \chi(n^2) = \chi(n)$ (using that χ is a completely multiplicative character of order dividing $2 \cdot 3^{K-1}$), giving $\Psi_\chi((n)) = \chi(n)$ as required. $\square$

Remark 2.6. The equality $\mathfrak{f}(\Psi_\chi) = \mathfrak{p}^{2K}$ with $\mathfrak{p}^{2K} = (3^K)$ is the geometric heart of the lifting: the ramification $(3) = \mathfrak{p}^2$ is exactly what converts conductor 3^K in $\mathbb{Q}$ to conductor $\mathfrak{p}^{2K}$ in $\mathcal{K}$, with no discrepancy. For unramified primes $\ell \neq 3$, the conductor of the lifted character in $\mathcal{K}$ would be ℓ^K (inert) or $\mathfrak{p}^K \bar{\mathfrak{p}}^K$ (split), neither of which matches ℓ^K exactly.

3. STEP 2: THE STARK–BAKER THEOREM FOR CM FIELDS

3.1. The CM field structure.

Proposition 3.1 (CM field properties of $\mathbb{Q}(\omega)$). *The field $\mathcal{K} = \mathbb{Q}(\omega)$ satisfies:*

- (i) $\mathcal{K}$ is a CM field (Proposition 2.1).
- (ii) $h_{\mathcal{K}} = 1$: the ring $\mathbb{Z}[\omega]$ is a PID (Proposition 2.1).
- (iii) $\text{disc}(\mathcal{K}) = -3$: the discriminant is -3 , so $|\text{disc}(\mathcal{K})| = 3$.
- (iv) $\mathcal{K}$ is one of the nine imaginary quadratic fields of class number 1 [8, 1].

Proof. (i)–(ii) are Proposition 2.1. (iii): The discriminant of $\mathbb{Q}(\sqrt{-3})$ is -3 (since $-3 \equiv 1 \pmod{4}$), the ring of integers is $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}] = \mathbb{Z}[\omega]$, giving discriminant -3 . (iv): The nine imaginary quadratic fields of class number 1 are $\mathbb{Q}(\sqrt{-d})$ for

$$d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\};$$

$d = 3$ appears on this list. □

3.2. The Stark–Baker theorem.

Theorem 3.2 (Stark–Baker [8, 1, 4]). *Let F be a CM field and let Ψ be a Hecke character of F of conductor $\mathfrak{f}$. Then $L(s, \Psi)$ has no real zero in the region*

$$\sigma > 1 - \frac{c_F}{\log(N(\mathfrak{f}) \cdot |\text{disc}(F)|)}$$

for an effectively computable constant $c_F > 0$ depending only on $[F : \mathbb{Q}]$.

Proof sketch. The CM condition gives F a complex conjugation ι such that $\iota \neq \text{id}$ and $F^+ = F^{(\iota)}$ is totally real. The Hecke L -function $L(s, \Psi)$ satisfies a functional equation of the form

$$\Lambda(s, \Psi) = W(\Psi) \cdot \Lambda(1-s, \bar{\Psi}),$$

where Λ is the completed L -function and $W(\Psi)$ is the root number. For CM fields, the functional equation relates $L(s, \Psi)$ to $L(1-s, \Psi)$ (since $\bar{\Psi} = \Psi \circ \iota$ and ι acts as complex conjugation).

A real zero $\beta \in (0, 1)$ of $L(s, \Psi)$ would force, via the functional equation, a zero of $L(s, \bar{\Psi})$ at $1 - \beta$. By Baker’s theorem on linear forms in logarithms [1], applied via Stark’s method [8], the product $L(\beta, \Psi) \cdot L(\beta, \chi_{-3})$ (where χ_{-3} is the quadratic character of $\mathcal{K}/\mathbb{Q}$) is bounded away from zero in the effective zero-free region. This gives the stated bound on c_F , with effectivity coming from Baker’s explicit lower bounds on linear forms in logarithms of algebraic numbers. See [5], Chapter 11, for the complete argument; the case $F = \mathbb{Q}(\sqrt{-3})$ is one of the nine cases resolved by Stark and Baker. □

Remark 3.3 (Scope and effectivity). The critical feature is *effectivity*: Baker’s theorem gives an explicit lower bound for linear forms in logarithms, making c_F computable. For imaginary quadratic fields (degree $[F : \mathbb{Q}] = 2$), the constant c_F is the same for all nine class-number-1 fields. For non-CM fields, Siegel zeros remain possible unconditionally: the CM condition is essential to exclude them.

3.3. Application to Ψ_χ .

Corollary 3.4 (Zero-free region for Ψ_χ). *The Hecke L -function $L(s, \Psi_\chi)$ has no real zero in*

$$\sigma > 1 - \frac{c'}{K \log 3} \sim 1 - \frac{c''}{\log q}$$

for effectively computable absolute constants $c', c'' > 0$.

Proof. Apply Theorem 3.2 with $F = \mathcal{K} = \mathbb{Q}(\omega)$, $\Psi = \Psi_\chi$, and $\mathfrak{f} = \mathfrak{p}^{2K}$. The conductor norm is $N(\mathfrak{p}^{2K}) = N(\mathfrak{p})^{2K} = 3^{2K}$. The discriminant is $|\text{disc}(\mathcal{K})| = 3$. Therefore:

$$\log(N(\mathfrak{f}) \cdot |\text{disc}(\mathcal{K})|) = \log(3^{2K} \cdot 3) = (2K + 1) \log 3 \sim 2K \log 3 = 2 \log q.$$

Theorem 3.2 gives the zero-free region $\sigma > 1 - c_K/(2 \log q)$, which is $\sigma > 1 - c''/\log q$ with $c'' = c_K/2$. $\square$

4. STEP 3: TRANSFERRING THE ZERO-FREE REGION

4.1. The L -function factorisation.

Lemma 4.1 (L -function factorisation). *Let χ be a primitive Dirichlet character modulo 3^K and let $\varepsilon = \left(\frac{\cdot}{3}\right)$ be the Kronecker symbol (the quadratic character of $\mathcal{K}/\mathbb{Q}$, of conductor 3). Then*

$$L(s, \Psi_\chi) = L(s, \chi) \cdot L(s, \chi\varepsilon).$$

Proof. This is the standard base-change factorisation for a Dirichlet character χ lifted to an imaginary quadratic field $\mathcal{K}$. We verify it via Euler products.

For a rational prime $\ell \neq 3$, there are two cases.

Case 1: $\ell \equiv 1 \pmod{3}$ (splits in $\mathcal{K}$). Write $\ell\mathbb{Z}[\omega] = \mathfrak{q}_1\mathfrak{q}_2$ with distinct primes $\mathfrak{q}_1 \neq \mathfrak{q}_2$. The local factor of $L(s, \Psi_\chi)$ at ℓ is $(1 - \Psi_\chi(\mathfrak{q}_1)\ell^{-s})^{-1}(1 - \Psi_\chi(\mathfrak{q}_2)\ell^{-s})^{-1}$. Since ℓ splits, $\varepsilon(\ell) = 1$, and the two local factors of $L(s, \chi)$ and $L(s, \chi\varepsilon)$ at ℓ multiply to give the same product.

Case 2: $\ell \equiv 2 \pmod{3}$ (stays inert in $\mathcal{K}$). Then $\ell\mathbb{Z}[\omega] = \mathfrak{q}$ is a prime of norm ℓ^2 . The local factor of $L(s, \Psi_\chi)$ at $\mathfrak{q}$ is $(1 - \Psi_\chi(\mathfrak{q})\ell^{-2s})^{-1}$. Since $\varepsilon(\ell) = -1$, the product of local factors of $L(s, \chi)$ and $L(s, \chi\varepsilon)$ at ℓ is $(1 - \chi(\ell)\ell^{-s})^{-1}(1 + \chi(\ell)\ell^{-s})^{-1} = (1 - \chi(\ell)^2\ell^{-2s})^{-1}$, which matches $L(s, \Psi_\chi)$ at ℓ since $\Psi_\chi(\mathfrak{q}) = \chi(\ell^2 \bmod 3^K) = \chi(\ell)^2$.

The ramified prime $\ell = 3$. Both sides have a conductor term at 3 that matches by construction of Ψ_χ (Proposition 2.5).

Special case $K = 1$. For $q = 3$, the only non-principal character modulo 3 is ε itself. Then $\chi = \varepsilon$, $\chi\varepsilon = \varepsilon^2 = \chi^0$ (principal), and $L(s, \chi\varepsilon) = \zeta(s)$ up to the Euler factor at 3. The factorisation $L(s, \Psi_\varepsilon) = L(s, \varepsilon) \cdot \zeta(s) = \zeta_{\mathcal{K}}(s)$ is the standard Dedekind factorisation.

Taking the Euler product over all primes gives the stated identity. $\square$

Remark 4.2. The twist $\chi\varepsilon$ is a Dirichlet character of conductor $\text{lcm}(3^K, 3) = 3^K$ for $K \geq 1$. It is therefore a character modulo 3^K , and the same Corollary 3.4 applies to $\Psi_{\chi\varepsilon}$, giving $L(s, \chi\varepsilon)$ a zero-free region of the same quality.

4.2. Completing the proof of Theorem 1.1.

Proof of Theorem 1.1. Let χ be a primitive character modulo $q = 3^K$. Suppose for contradiction that $L(s, \chi)$ has a real zero at $\beta > 1 - c/\log q$ for some $c > 0$.

Step 1. By Proposition 2.5, χ lifts to a primitive Hecke character Ψ_χ of $\mathcal{K}$ of conductor $\mathfrak{p}^{2K}$.

Step 2. By Lemma 4.1, $L(\beta, \Psi_\chi) = L(\beta, \chi) \cdot L(\beta, \chi\varepsilon)$. Since $L(\beta, \chi) = 0$ by hypothesis, we have $L(\beta, \Psi_\chi) = 0$ (note: $L(\beta, \chi\varepsilon)$ has no pole at β , since $\beta < 1$ and poles of Dirichlet L -functions occur only at $s = 1$ for principal characters; $\chi\varepsilon$ is non-principal for all $K \geq 1$ since χ has conductor $3^K \geq 3$ while ε has conductor 3, and $\chi\varepsilon \neq \chi^0$).

Step 3. By Corollary 3.4, $L(s, \Psi_\chi)$ has no real zero in $\sigma > 1 - c''/\log q$. But $\beta > 1 - c/\log q$, so for $c < c''$ we have β in the zero-free region of $L(s, \Psi_\chi)$. This contradicts $L(\beta, \Psi_\chi) = 0$.

Conclusion. No such β exists. Taking $c = c''/2$ (which is effective since c'' is effective by Remark 3.3), we have the stated zero-free region for $L(s, \chi)$.

Watt's condition $\theta_f \leq 7/32$. The complete absence of Siegel zeros in $\sigma > 1 - c/\log q$, combined with the standard zero-density estimate for Dirichlet L -functions of the form $N(\sigma, T, \chi) \ll (qT)^{4(1-\sigma)/(3-2\sigma)+\varepsilon}$ (see [5], Theorem 10.4), gives Watt's condition $\theta_f \leq 7/32$ for the character family $\{\chi \pmod{3^K}\}$. $\square$

5. WHY THIS ARGUMENT IS SPECIFIC TO $q = 3^K$

The proof uses three properties of 3 in $\mathbb{Z}[\omega]$ simultaneously. We show that no other prime-power family satisfies all three.

Proposition 5.1 (Specificity of the CM argument). *The lifting of Proposition 2.5 and the Stark–Baker application of Section 3 work for $q = 3^K$ because:*

- (1) Ramification: 3 is the unique rational prime that ramifies in $\mathcal{K} = \mathbb{Q}(\sqrt{-3})$, so the conductor match $\mathfrak{f}(\Psi_\chi) = \mathfrak{p}^{2K} = (3^K)$ holds exactly.
- (2) Cyclic unit group: $(\mathbb{Z}/3^K\mathbb{Z})^\times \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ is a single cyclic group for all $K \geq 1$, enabling the surjectivity of the norm map in Proposition 2.5.
- (3) Unique CM field: $\mathcal{K} = \mathbb{Q}(\sqrt{-3})$ is the unique imaginary quadratic field that is ramified only at 3, so characters of $(\mathbb{Z}/3^K\mathbb{Z})^\times$ correspond precisely to Hecke characters of $\mathcal{K}$ of conductor $\mathfrak{p}^{2K}$.

No other prime-power family $q = \ell^K$ with $\ell \neq 3$ satisfies all three properties simultaneously.

Proof. Properties (1)–(3) for $\ell = 3$ follow from Propositions 2.1 and 2.2 and Lemma 3.1 of [15].

For other primes $\ell \neq 3$:

$\ell \equiv 1 \pmod{3}$ (*splits in $\mathcal{K}$*). Then ℓ does not ramify in $\mathcal{K}$, so property (1) fails: the conductor of the lifted character in $\mathcal{K}$ would be $\ell^K \bar{\ell}^K$ (a product of two distinct prime ideals), not matching ℓ^K as a single ideal.

$\ell \equiv 2 \pmod{3}$ (*inert in $\mathcal{K}$*). Then ℓ remains prime in $\mathcal{K}$. A Hecke character of conductor ℓ^K (as an ideal of $\mathbb{Z}[\omega]$) would correspond to a character of $(\mathbb{Z}[\omega]/(\ell^K))^\times$, which has order $\ell^{2K-2}(\ell^2 - 1)$ —not matching $(\mathbb{Z}/\ell^K\mathbb{Z})^\times$ of order $\ell^{K-1}(\ell - 1)$. Property (1) fails.

$\ell = 2$. The prime 2 splits in $\mathcal{K}$ (since $-3 \equiv 1 \pmod{8}$), so property (1) fails.

In all cases, either the conductor does not match ($\ell \neq 3$), or the relevant CM field is not $\mathbb{Q}(\omega)$ and a separate argument would be required for each ℓ . $\square$

Remark 5.2. For the purposes of the companion paper [15], only $\ell = 3$ is needed. Proposition 5.1 explains *why* the argument is specific to this case and rules out simple generalisation to other families.

6. FULL COHERENCE: ONE STRUCTURE, THREE APPLICATIONS

The companion paper [15] establishes $\theta_W = 5/8$ for cascade moduli using three distinct applications of the Eisenstein integer structure. It is instructive to see that they all draw on the same algebraic fact.

Proposition 6.1 (Coherence of the Eisenstein proof). *The proof of Theorem 1.1 of [15] uses the ramification $(3) = \mathfrak{p}^2$ in $\mathbb{Z}[\omega]$ for three separate purposes, each essential:*

- (1) **Part I (Filtration).** *The ring $\mathbb{Z}[\omega]/(3^K)$ carries a natural filtration with three independent levels, induced by $(3^K) = \mathfrak{p}^{2K}$. This forces sieve weights over 3^K to be triply-well-factorable, enabling three-fold Cauchy–Schwarz and giving ceiling $\theta \leq 2/3$.*
- (2) **Part II (Cyclic cancellation).** *The unit group $(\mathbb{Z}/3^K\mathbb{Z})^\times \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ is cyclic (a consequence of the prime-power structure enforced by ramification). This enables Kloosterman sum cancellation beyond the Weil bound, improving the ceiling to $\theta = 5/8$.*
- (3) **Condition W3 (This paper).** *The ramification $(3) = \mathfrak{p}^2$ makes the conductor of the lifted Hecke character Ψ_χ match the conductor of χ exactly (Remark 2.6), enabling the Stark–Baker theorem to be applied to $L(s, \Psi_\chi)$ and the zero-free region to be transferred to $L(s, \chi)$.*

The same geometric fact—ramification of 3 in the hexagonal lattice $\mathbb{Z}[\omega]$ —does the work in all three applications.

This coherence is not a coincidence. It reflects that cascade moduli $q = 3^K$ are the *natural* moduli for the Eisenstein integer framework: they are the moduli for which the algebraic geometry of $\mathbb{Z}[\omega]$ can be deployed systematically. The level of distribution $\theta_W = 5/8$ is not a computational accident; it is forced by the hexagonal lattice.

7. CLOSING THE GAP: UNCONDITIONAL STATEMENT

We collect the complete verification of Watt’s conditions and state the unconditional form of the main theorem of [15].

Theorem 7.1 (Watt conditions: complete verification). *For cascade moduli $q = 3^K \leq Q = x^{5/8-\varepsilon}$, all three conditions of Watt’s theorem [10] hold:*

Condition	Statement	Source	Status
W1	$M \leq q$	[15], App. A	✓
W2	$MR \leq q^2$	[15], App. A	✓
W3	No Siegel zeros	Theorem 1.1 above	✓

Corollary 7.2 (Unconditional level of distribution). *Theorem 1.1 of [15]—the level of distribution $\theta_W = 5/8$ for cascade moduli—holds unconditionally. The annotation [NOTE] in Appendix A of [15] is resolved.*

Proof. All three Watt conditions are now verified (Theorem 7.1). Watt’s theorem therefore applies to cascade moduli 3^K , and the proof of Theorem 1.1 of [15] goes through without further hypotheses. $\square$

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THE ARC INSTITUTE OF KNOWWARE, CALGARY, ALBERTA, CANADA
Email address: `the@knowware.institute`